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Department of Computer Science & Engineering

Module 1:

Chapter 2: Understanding of Data

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What is Data?

- Data are facts
- Facts are in the form of **numbers, audio, video, and image**
- Today, **business organizations are accumulating vast and growing amounts of data** of the order of gigabytes, tera bytes, exabytes.
- Data is available in different data sources like **flat files, databases, or data warehouses.**
- **Need to analyze data for taking decisions**

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Characteristics of Big Data

- **Volume:** Since there is a reduction in the cost of storing devices, there has been a tremendous growth of data.
- Big Data is measured in terms of petabytes (PB) and exabytes (EB). One exabyte is 1 million terabytes.
- **Velocity:** The fast arrival speed of data and its increase in data volume is noted as velocity.
- **Variety :** The variety of Big Data includes:
  - **Form:** There are many forms of data. Data types range from text, graph, audio, video, to maps.
  - **Function:** These are data from various sources like human conversations, transaction records, and old archive data.
  - **Source of data: T**he data source can be classified as open/public data, social media data and multimodal data.
- **Validity:** Validity is the *accuracy of the data for taking decisions*
- **Value:** Value is the characteristic of big data that indicates *the value of the information that is extracted from the data.*

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Types of Data

- In Big Data, there are three kinds of data.
- Structured data
- Unstructured data
- Semi-structured data

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Structured data

- In structured data, data is stored in an organized manner such as a **database** where it is available in the form of a **table**.
- The **data can also be retrieved** in an organized manner using tools like **SQL.**
- **Record data:** We have a collection of objects in a dataset and each object has a **set of measurements.**
  - The measurements can be arranged in the form of a matrix.
  - **Rows in the matrix represent an object** and can be called as **entities, cases, or records.**
  - The **columns** of the dataset are called **attributes, features, or fields.**
- **Graphics data:** It involves the *relationships among objects.*
  - For example, a web page can refer to another web page.
  - This can be modeled as a graph.
  - The nodes are web pages and the hyperlink is an edge that connects the nodes.

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Structured data (Contd…)

- **Data matrix:** It is a variation of the record type because *it consists of numeric attributes.*
  - ✓ The standard matrix operations can be applied on these data.
- **Ordered data** :Ordered data objects involve attributes that have an implicit order among them.
  - **Temporal data :** It is the **data whose attributes are associated with time.** E.g: customer purchasing patterns during festival time
  - **Sequence data:** sequential data but **does not have time stamps**, This data involves the **sequence of words or letters.**
  - **Spatial data**: It has attributes such as **positions or areas.** For example, maps are spatial data where the points are related by location.

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Unstructured data

- Video, image, programs
- Blog data
- 80% of the data is unstructured

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Semi-structured data

- Semi-structured data are partially structured and partially unstructured
- XML/JSON objects
- RSS (Really Simple Syndication) feeds
- Hierarchical records

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Data Storage: Flat Files

- Approaches to organize and manage data in **storage files and systems from flat file to data warehouses.**
- **Flat Files:** These are the simplest and most commonly available data source.
- It is also the cheapest way of organizing the data.
- These flat files are the files where data is stored in plain **ASCII format.**
- Minor changes of data in flat files affect the results of the data mining algorithms.
- Flat file is suitable only for storing small dataset.
- Some of the popular spreadsheet formats are listed below:
  - CSV files - CSV stands for comma-separated value files
  - TSV files - TSV stands for Tab separated values files where values are separated by Tab.
- There are many tools like Google Sheets and Microsoft Excel to process these files.

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Data Storage: Database System

- It normally consists of database files and a database management system (DBMS).
- Database files contain original data and metadata.
- DBMS aims to manage data and improve operator performance .
- A relational database consists of sets of tables.
- The tables have rows and columns.
- The columns represent the attributes and rows represent tuples.
- A tuple corresponds to either an object or a relationship between objects.
- A user can access and manipulate the data in the database using SQL.
- Different types of databases
  - A transactional database
  - Time-series database : Log Files

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Data Storage: Other Types

- **World Wide Web (WWW):** It provides a diverse, worldwide online information source.
  - The objective of data mining algorithms is to mine interesting patterns of information present in WWW
- **XML (eXtensible Markup Language):** It is both human and machine interpretable data format that can be *used to represent data that needs to be shared across the platforms.*
- **Data Stream** It is dynamic data, which flows in and out of the observing environment.
  - Typical characteristics of data stream are **huge volume of data, dynamic, fixed order movement, and real-time constraints.**
- **RSS (Really Simple Syndication)** It is a format for *sharing instant feeds* across services.
- **JSON (JavaScript Object Notation)** It is another useful *data interchange format that is often used for many machine learning algorithms.*

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BIG DATA ANALYTICS

- The primary aim of data analysis is to **assist business organizations to take decisions.**
- For example, a business organization may want to know which is the *fastest selling product,* in order for them to market activities.
- Data analysis and data analytics are terms that are used interchangeably to refer to the same concept.
- Data analytics is a general term and data analysis is a part of it.
- **Data analytics refers to the process of data collection, preprocessing and analysis.**
- **Data analysis is just analysis and is a part of data analytics.**
- Data analytics, instead, *concentrates more on future and helps in prediction.*

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TYPES OF ANALYTICS

- There are four types of data analytics:

1. Descriptive analytics
2. Diagnostic analytics
3. Predictive analytics
4. Prescriptive analytics

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Descriptive Analytics

- It is about **describing the main features of the data.**
- It is often stated that **analytics is essentially statistics.**
- There are two aspects of statistics - Descriptive and Inference.
- Descriptive analytics only focuses on the description part of the data.
- Inference is the process of drawing conclusions by reasoning, logic and analytical thinking.

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Diagnostic Analytics

- It deals with the question - 'Why?'.
- This is also known as causal analysis, as it aims to **find out the cause and effect of the events.**
- For example, if a product is not selling, diagnostic analytics aims to find out the reason.

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Predictive Analytics

- It deals with the future.
- It deals with the question — 'What will happen in future given this data?'.
- This involves the **application of algorithms to identify the patterns to predict the future.**
- The entire course of machine learning is mostly about predictive analytics

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Prescriptive Analytics

- Prescriptive analytics goes beyond prediction and **helps in decision making by giving a set of actions.**
- It helps the organizations to plan better for the future and to mitigate the risks that are involved.

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BIG DATA ANALYSIS FRAMEWORK

- Big data framework is a layered architecture.
- A 4-layer architecture has the following layers:

1. Data connection layer
2. Data management layer
3. Data analytics later
4. Presentation layer

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Data Connection Layer

- It has **data ingestion mechanisms and data connectors.**
- **Data ingestion** means taking raw data and importing it into appropriate data structures.
- It performs the tasks of ETL process.
- By ETL, it means extract, transform and load operations.

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Data Management Layer

- It performs **preprocessing of data**.
- The purpose of this layer is to allow parallel execution of queries, and read, write and data management tasks.
- There may be many schemes that can be implemented by this layer such as **data-in-place, where the data is not moved at all,** or constructing data repositories such as data warehouses and pull data on-demand mechanisms.

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Data Analytic Layer

- It has many functionalities such as **statistical tests, machine learning algorithms to understand, and construction of machine learning models.**
- Types of Processing:
  - **Cloud Computing:** IaaS, PaaS, SaaS, public, private, hybrid cloud
  - **Grid Computing:** Parallel and distributed computing framework
  - **H-Computing or HPC**

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Presentation Layer

- It has mechanisms such as **dashboards, and applications** that *display the results of analytical engines and machine learning algorithms.*
- **Big Data processing cycle** involves data management that consists of the following steps.
  – Data collection
  – Data preprocessing
  – Applications of machine learning algorithm
  – Interpretation of results and visualization of machine learning algorithm

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Data Collection

- The first task of **gathering datasets** are the collection of data.
- A good quality data yields a better result.
- Properties of 'Good Data':
  - *Timeliness:* The data should be relevant and not stale or obsolete data.
  - *Relevancy :* The data should be relevant and ready for the machine learning or data mining algorithms.
  - *Knowledge about the data:* The data should be understandable and interpretable
- Classification of data source:
  - *Open or public data source :* It is a data source that does not have any stringent copyright rules or restrictions. E.g: Government census data
  - *Social media:* Twitter, Facebook, YouTube, and Instagram.
  - *Multimodal data:* It includes data that involves many modes such as text, video, audio and mixed types.

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Data Preprocessing

- In real world, the available data is **'dirty'.**
- By this word 'dirty', it means:
  - Incomplete data
  - Inaccurate data
  - Outlier data
  - Data with missing values
  - Data with inconsistent value
  - Duplicate data
- Data preprocessing improves the quality of the data mining techniques.
- The raw data must be preprocessed to give accurate results.
- The process of detection and removal of errors in data is called **data cleaning.**
- **Data wrangling** means making the data processable for machine learning algorithms.

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Illustration of 'Bad' Data

Patient ID	Name	Age	Date of Birth (DoB)	Fever	Salary
1.	John	21		Low	-1500
1.	Andre	36		High	Yes
1.	David	5	10/10/1980	Low	” ”
1.	Raju	136		High	Yes

Outliers are data that exhibit the characteristics that are different from other data and have very unusual values.

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Missing Data Analysis

- Data cleaning process is missing data analysis.
- **Data cleaning routines attempt to fill up the missing values, smoothen the noise while identifying the outliers and correct the inconsistencies of the data.**
- This enables data mining to **avoid overfitting** of the models.

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Missing Data Analysis:
solve the problem of missing data

- **Ignore the tuple -** A tuple with missing data, especially the class label, is ignored.
- **Fill in the values manually -** Here, the domain expert can analyse the data tables and carry out the analysis and fill in the values manually.
- **A global constant** can be used to fill in the missing attributes. The missing values may be 'Unknown' or be 'Infinity'.
- The attribute value may be **filled by the attribute value**. Say, the average income can replace a missing value.
- **Use the attribute mean** for all samples belonging to the same class. Here, the average value replaces the missing values of all tuples that fall in this group.
- **Use the most possible value** to fill in the missing value. The most probable value can be obtained from other methods like classification and decision tree prediction.

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Removal of Noisy or Outlier Data

- **Noise is a random error or variance in a measured value.**
- **It can be removed by using binning,** which is a method where the given data values are sorted and distributed into equal frequency bins.
- The bins are also called as buckets.
- The binning method then uses the neighbor values to smooth the noisy data.
- **'smoothing by means'** where the mean of the bin removes the values of the bins
- **'smoothing by bin medians'** where the bin median replaces the bin values
- **'smoothing by bin boundaries'** where the bin value is replaced by the closest bin boundary. The maximum and minimum values are called bin boundaries.

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Example

- Consider the following set:

S = {12, 14, 19, 22, 24, 26, 28, 31, 34}. Apply various binning techniques and show the result.

- Solution: By equal-frequency bin method, the data should be distributed across bins.
- Let us assume the bins of size 3, then the above data is distributed across the bins

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Noisy Data

BINNING TECHNIQUE

$S = \{12, 14, 19, 22, 24, 26, 28, 31, 34\}$

Bin 1 : 12 , 14, 19

Bin 2 : 22, 24, 26

Bin 3 : 28, 31, 32

By smoothing bins method, the bins are replaced by the bin means. This method results in:

Bin 1 : 15, 15, 15

Bin 2 : 24, 24, 24

Bin 3 : 30.3, 30.3, 30.3

Using smoothing by bin boundaries method, the bins' values would be like:

Bin 1 : 12, 12, 19

Bin 2 : 22, 22, 26

Bin 3 : 28, 32, 32

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Data Integration and Data Transformations

- Data integration involves routines that **merge data from multiple sources into a single data source.**
- this may lead to redundant data.
- The main goal of data integration is to detect and remove redundancies that arise from integration.
- **Data transformation routines perform operations like normalization** to improve the performance of the data mining algorithms.
- In normalization, the attribute values are scaled to **fit in a range (say 0-1)**
- Some of the normalization procedures used are:
  - *Min-Max*
  - z-Score

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Min-Max Procedure

- It is a normalization technique where **each variable V is normalized by its difference with the minimum value divided by the range to a new range, say 0-1.**

$$min\text{-}max = \frac{V - min}{max - min} \times (new\ max - new\ min) + new\ min$$

- Here *max-min* is the range.
- *Min* and *max* are the minimum and maximum of the given data
- *new max* and *new min* are the minimum and maximum of the target range, say 0 and 1

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Min-Max Procedure : Example

- Consider the set: $V = \{88, 90, 92, 94\}$. Apply *Min-Max* procedure and map the marks to a new range 0-1.

$$min\text{-}max = \frac{V - min}{max - min} \times (new\ max - new\ min) + new\ min$$

For marks 88,

$$min\text{-}max = \frac{88 - 88}{94 - 88} \times (1 - 0) + 0 = 0$$

Similarly, other marks can be computed as follows:

For marks 90,

$$min\text{-}max = \frac{90 - 88}{94 - 88} \times (1 - 0) + 0 = 0.33$$

For marks 92,

$$min\text{-}max = \frac{92 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{4}{6} = 0.66$$

For marks 94,

$$min\text{-}max = \frac{94 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{6}{6} = 1$$

So, it can be observed that the marks {88, 90, 92, 94} are mapped to the new range {0, 0.33, 0.66, 1}.

Thus, the *Min-Max* normalization range is between 0 and 1.

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z-Score Normalization

- This procedure works by taking **the difference between the field value and mean value, and by scaling this difference by**

$$V* = V - \mu/\sigma \tag{2.2}$$

Here, σ is the standard deviation of the list V and μ is the mean of the list V.

Consider the mark list $V = \{10, 20, 30\}$, convert the marks to z-score.

Solution: The mean and Sample Standard deviation (σ) values of the list V are 20 and 10, respectively. So the z-scores of these marks are calculated using Eq. (2.2) as:

$$\text{z-score of } 10 = \frac{10-20}{10} = -\frac{10}{10} = -1$$

$$\text{z-score of } 20 = \frac{20-20}{10} = \frac{0}{10} = 0$$

$$\text{z-score of } 30 = \frac{30-20}{10} = \frac{10}{10} = 1$$

Hence, the z-score of the marks 10, 20, 30 are –1, 0 and 1, respectively.

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DESCRIPTIVE STATISTICS

- Descriptive statistics is a branch of statistics that **does dataset summarization.**
- It is used to summarize and describe data.
- **Data visualization** is a branch of study that is useful for investigating the given data.
- Mainly, the **plots** are useful to explain and present data to customers.
- Descriptive analytics and data visualization techniques help to understand the nature of the data,
- **Exploratory Data Analysis (EDA):** helps to determine the kinds of machine learning or data mining tasks that can be applied to the data

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Dataset and Data Types

- A dataset can be assumed to be a **collection of data objects.**
- The data objects may be records, points, vectors, patterns, events, cases, samples or observations.
- These records contain many attributes.
- An attribute can be defined as the property or characteristics of an object.
- The type of attribute determines the data types, often referred to as measurement scale types.

Table 2.2: Sample Patient Table

Patient ID	Name	Age	Blood Test	Fever	Disease
1.	John	21	Negative	Low	No
2.	Andre	36	Positive	High	Yes

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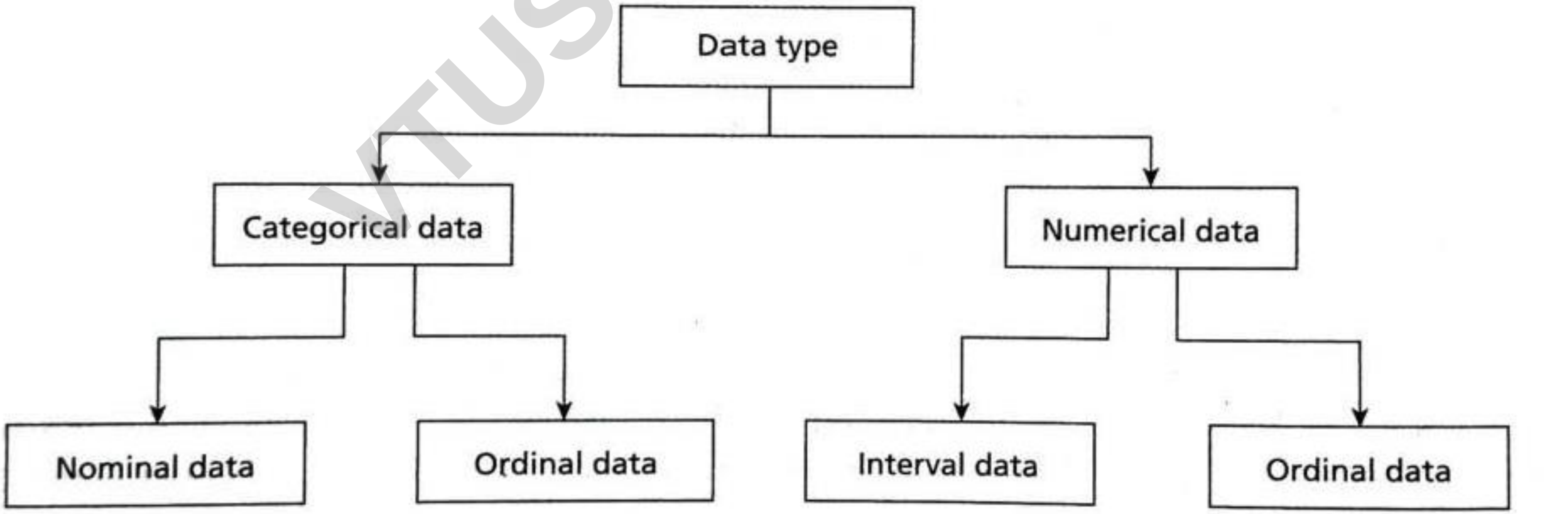
Types of data



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Categorical or Qualitative Data

- The categorical data can be divided into two types. They are nominal type and ordinal type.
  - **Nominal Data -** In Table 2.2, patient ID is nominal data. Nominal data are **symbols and cannot be processed like a number.**
  - Nominal data type provides only information but has no ordering among data.
  - Only operations like **(=, *)** are meaningful for these data.
  - **Ordinal Data -** It provides enough information and has natural order. **For example, Fever= (Low, Medium, High)** is an ordinal data.

Table 2.2: Sample Patient Table

Patient ID	Name	Age	Blood Test	Fever	Disease
1.	John	21	Negative	Low	No
2.	Andre	36	Positive	High	Yes

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Numeric or Quantitative Data

- It can be divided into two categories. They are interval type and ratio type.
  - **Interval Data -** Interval data is a numeric data for which the **differences between values** are meaningful.
  - For example, **there is a difference between 30 degree and 40 degree.**
  - **Ratio Data -** For ratio data, both differences and ratio are meaningful.
- **Discrete Data** This kind of data is recorded as integers. **Employee identification** number such as 10001 is discrete data.
- **Continuous Data** It can be **fitted into a range and includes decimal point.** For example, age is a continuous data.

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- classifying the data is based on the number of variables used in the dataset
- the data can be classified as **univariate data, bivariate data, and multivariate data**
- In case of **univariate data,** the dataset has only one variable. A variable is also called as category.
- Bivariate data indicates that the number of variables used are two and multivariate data uses three or more variables.

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UNIVARIATE DATA ANALYSIS AND VISUALIZATION

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- Univariate analysis is the **simplest form of statistical analysis.**
- As the name indicates, the **dataset has only one variable.**
- A variable can be called as a **category.**
- Univariate does not deal with cause or relationships.
- The aim of **univariate analysis is to describe data and find patterns.**
- Univariate data description involves **finding the frequency distributions, central tendency measures, dispersion or variation, and shape of the data.**

Data Visualization

- Data visualization **helps to understand data.**
- It helps to present information and data to customers.
- Some of the g**raphs that are used in univariate data analysis** are:
  - bar charts,
  - histograms,
  - frequency polygons and
  - pie charts.

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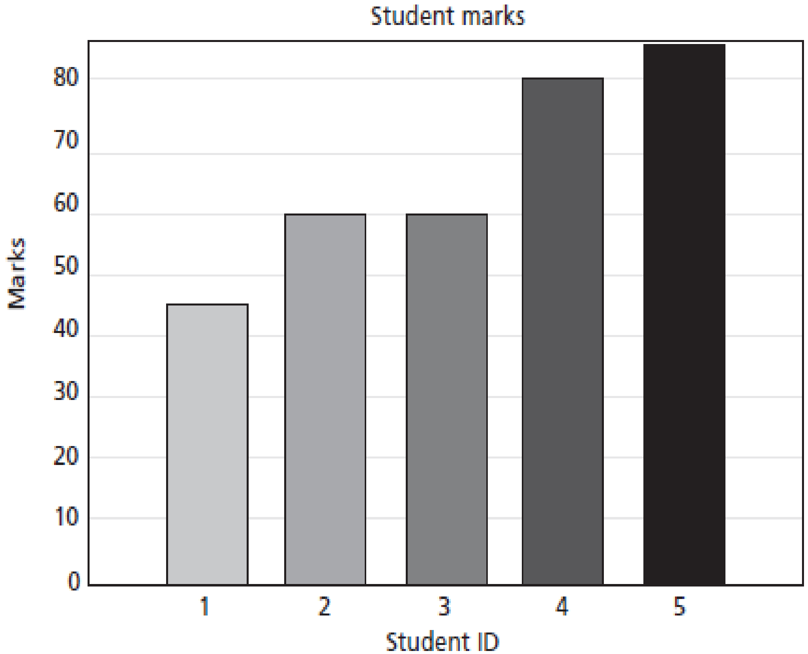
Bar Chart

- A Bar chart (or Bar graph) is used to display the **frequency distribution for variables.**
- Bar charts are used to **illustrate discrete data.**
- The charts can also **help to explain the counts of nominal data.**
- It also helps in **comparing the frequency of different groups.**



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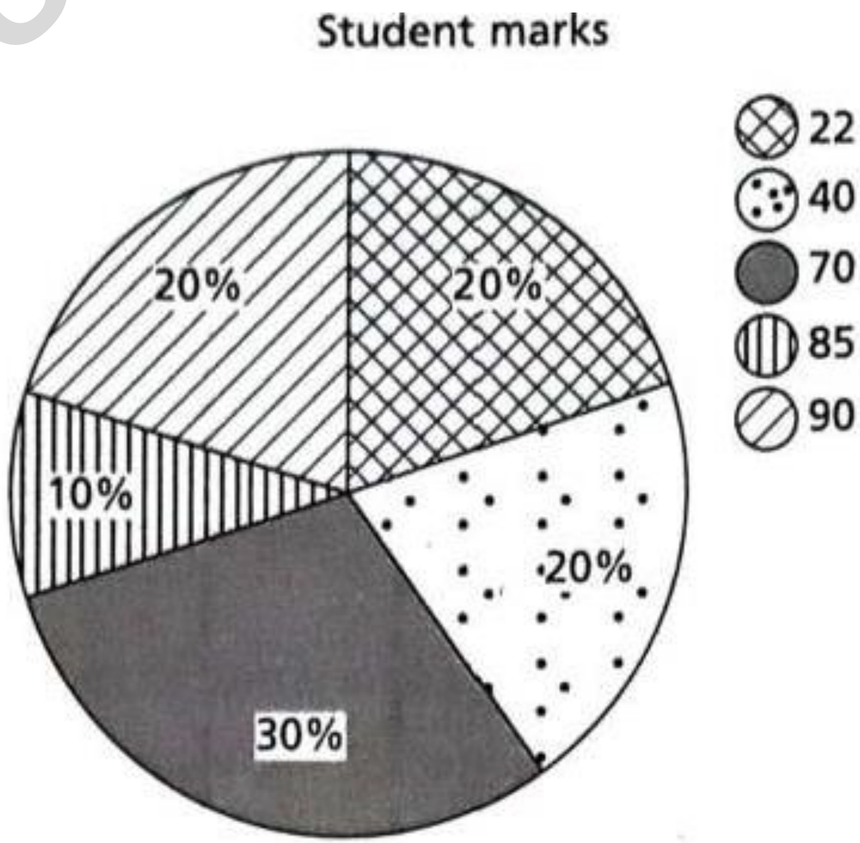
Pie Chart

- These are equally helpful in illustrating the univariate data.
- The **percentage frequency distribution** of students' marks [122, 22, 40, 40, 70, 70, 70, 85, 90, 90]



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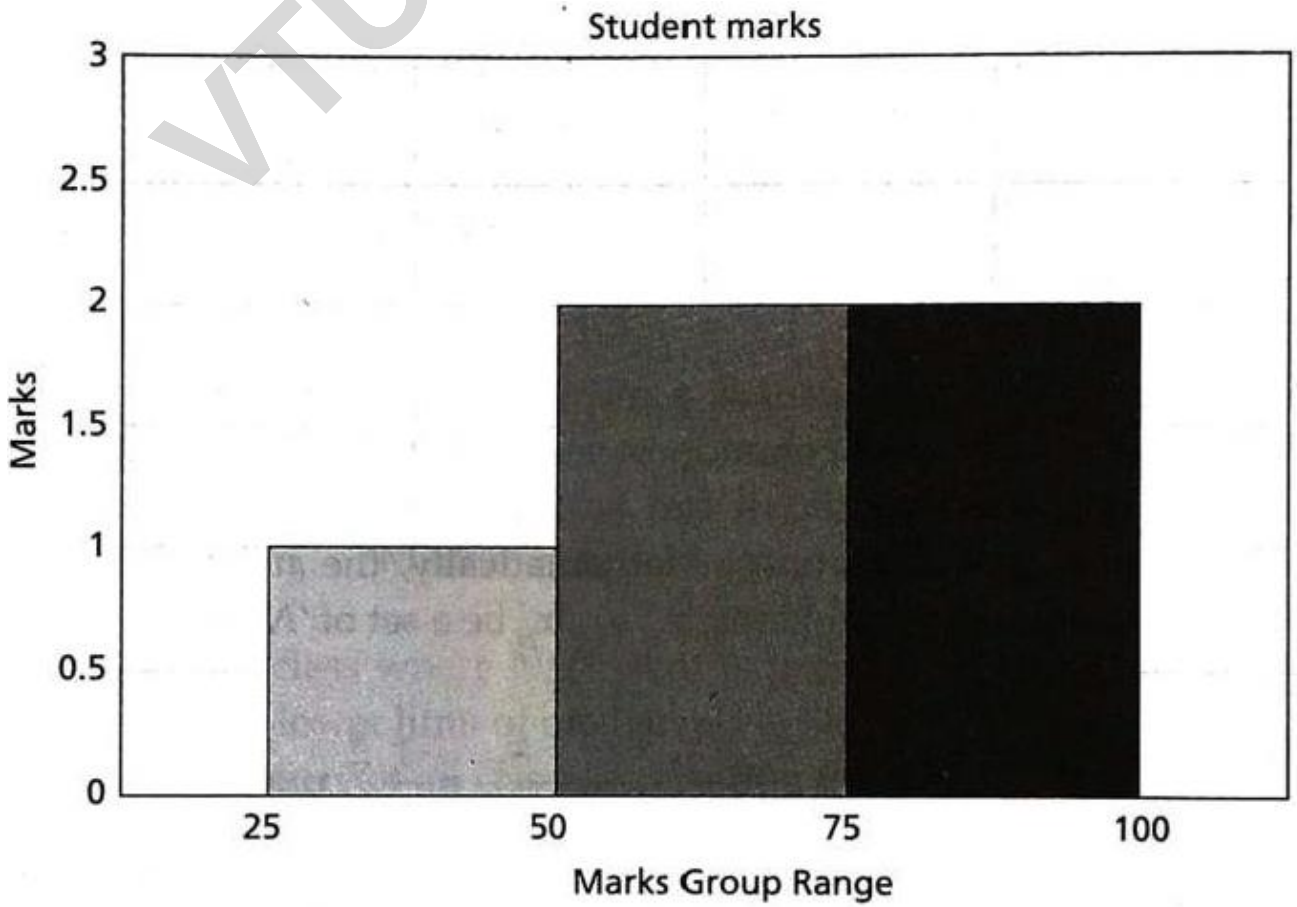
Histogram

- It plays an important role in data mining for showing **frequency distributions.**
- Histograms can be used as charts to **show frequency, skewness present in the data, and shape./ Mode(peak)**
- The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0-25, 26-50, 51-75, 76-100 is



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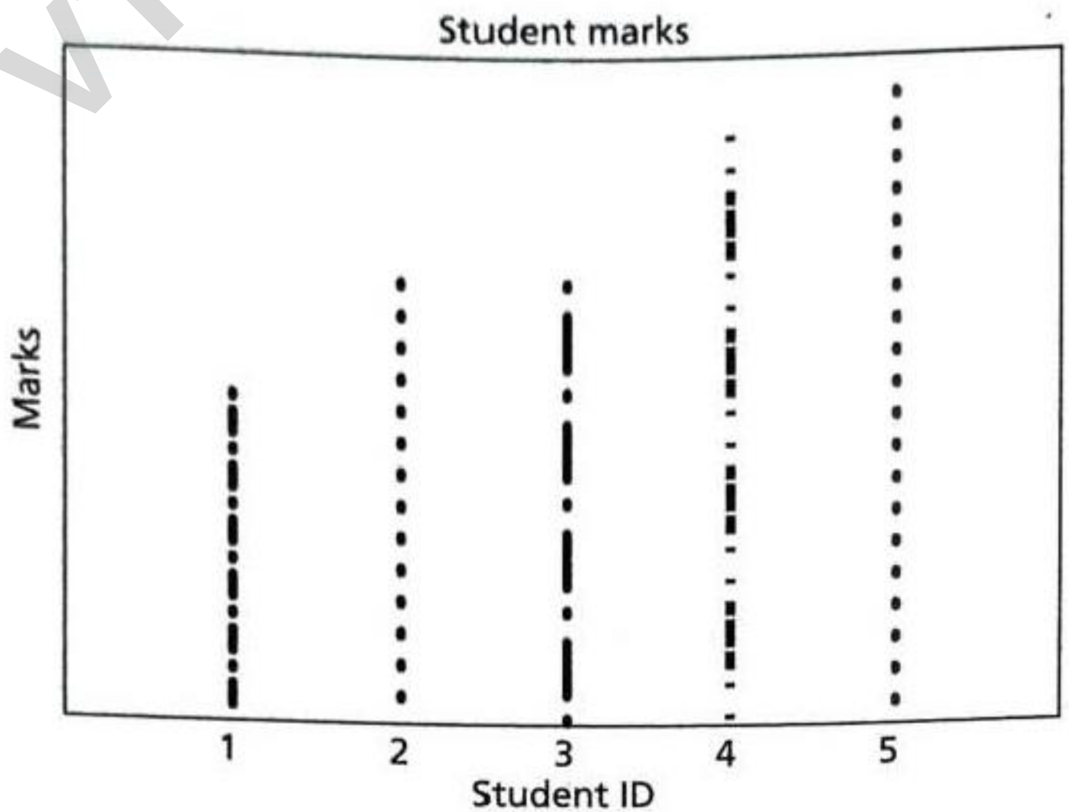
Dot Plots

- These are similar to bar charts.
- They are less clustered as compared to bar charts, as they **illustrate the bars only with single points**



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Central Tendency

- **Summary of the data is called central tendency.**
- Central tendency can explain the characteristics of data and that further helps in comparison.
- Mass data have tendency to concentrate at certain values, normally in the central location. It is called **measure of central tendency (or averages).**
- Popular measures are **mean, median and mode.**

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Central Tendency-Mean

- **Mean -** Arithmetic average (or mean) is a measure of central tendency that **represents the 'center' of the dataset.**
- E.g: average income or average traffic.

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

For example, the mean of the three numbers 10, 20, and 30 is $\frac{10+20+30}{3} = \frac{60}{3} = 20$

- **Weighted mean -** Unlike arithmetic mean that gives the weightage of all items equally, **weighted mean gives different importance to all items** as the item importance varies.

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Central Tendency-Mean

- **Geometric mean -** Let x_1, $x2$, ••• x_N be a set of 'N' values or observations. **Geometric mean is the Nth root of the product of *N* items.**
- The formula for computing geometric mean is given as follows:

$$\text{Geometric mean} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \cdots \times x_N} \tag{2.4}$$

Here, n is the number of items and x_i are values. For example, if the values are 6 and 8, the geometric mean is given as $\sqrt[2]{6 \times 8} = \sqrt{48}$. In larger cases, computing geometric mean is difficult. Hence, it is usually calculated as:

$$\text{Anti-log of } \frac{\log(x_1) + \log(x_2) + \cdots + \log(x_N)}{N} \tag{2.5}$$

$$= \text{anti-log } \frac{\sum_{i=1}^{n} \log(x_i)}{N} \tag{2.6}$$

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Central Tendency-Meadian

- **Median -** The middle value in the distribution is called median.
- If the total number of items in the distribution is odd, then the middle value is called median.
- If the numbers are even, then the average value of two items in the centre is the median.

$$\text{Median} = L_1 + \frac{\frac{N}{2} - cf}{f} \times i \qquad (2.7)$$

Median class is that class where N/2[th] item is present. Here, i is the class interval of the median class and L_1 is the lower limit of median class, f is the frequency of the median class, and cf is the cumulative frequency of all classes preceding median.

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Central Tendency-Mode

- **Mode -** Mode is the **value that occurs more frequently in the dataset.**
- In other words, the value that has the **highest frequency is called mode.**
- The procedure for finding the mode is to calculate the frequencies for an the values in the data, and mode is the value (or values) with the highest frequency.
- Normally, the dataset is classified as **unimodal, bimodal and trimodal with modes 1, 2 and 3, respectively.**

Dispersion

- The **spreadout of a set of data around the central tendency (mean, median or mode) is called dispersion.**
- Dispersion is represented by various ways such as **range, variance, standard deviation, and standard error.**
- **Range** Range is the difference between the maximum and minimum of values of the given list of data.
- **Standard deviation** is the average distance from the mean of the dataset to each point.

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}} \tag{2.8}$$

Here, N is the size of the population, x_i is observation or value from the population and μ is the population mean. Often, $N-1$ is used instead of N in the denominator of Eq. (2.8). The reason is that for larger real-world, the division by $N-1$ gives an answer closer to the actual value.

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Quartiles and Inter Quartile Range

- **Percentiles are about data that are less than the coordinates by some percentage of the total value.**
- **K^{th} percentile is the property that the *k%* of the data lies at or below X_i .**
- median is 50^{th} percentile and can be denoted as $Q_{0.50}$. (Q2)
- The 25^{th} percentile is called first quartile (Q_1)
- The 75th percentile is called third quartile (Q_3).
- Another measure that is useful to measure dispersion is **Inter Quartile Range (IQR).**
- The IQR is the difference between Q_3 and Q_1.
- Interquartile percentile = Q_3 — *Q1*
- **Outliers** are normally the values falling apart at least by the amount 1.5 x IQR above the third quartile or below the first quartile.

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Quartiles and Inter Quartile Range

- E.g: For patients' age list {12, 14, 19, 22, 24, 26, 28, 31, 34}, find the IQR.
- The median is in the fifth position. In this case, 24 is the median.
- The **first quartile** is median of the scores below the mean i.e., {12, 14, 19, 22}.
- In this case, the median is the average of the second and third values, that is, $\mathbf{Q_{0.25} = 16.5}$**.**
- Similarly, the **third quartile** is the median of the values above the median, that is {26, 28, 31, 34}
- $\mathbf{Q_{0.75}}$ is the average of the seventh and eighth score. In this case, it is 28 + 31/2 = 59/2 = **29.5.**
- IQR= $Q_{0.75}$ — $Q_{0.25}$= *29.5-16.5=13*

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Five-point Summary and Box Plots

- The median, quartiles Q_1 and Q_y and minimum and maximum written in the **order < Minimum, Q_1, Median, Q_3 Maximum >** is known as five-point summary.
- **Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9}.**
- The minimum is 2 and the maximum is 13.
- The Q_1, Q_2 and Q_3 are 3, 8 and 11, respectively.
- 5-point summary is {2, 3, 8, 11, 13}

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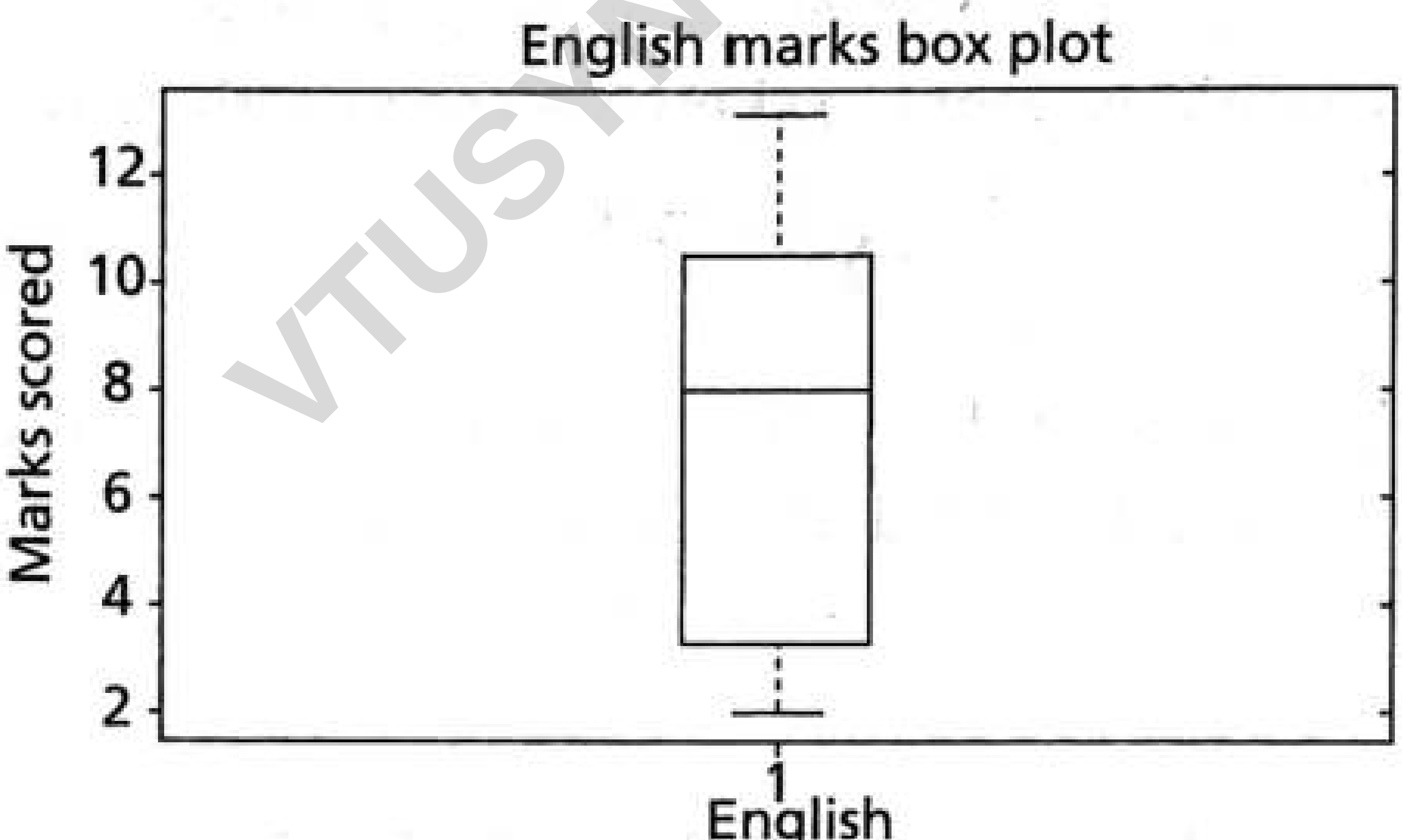
5-point summary is {2, 3, 8, 11, 13} : Box Plot



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Shape

- **Skewness and Kurtosis (called moments)** indicate the symmetry/asymmetry and peak location of the dataset.
- **Ideally, skewness should be zero as in ideal normal distribution.**
- **More often, the given dataset may not have perfect symmetry**

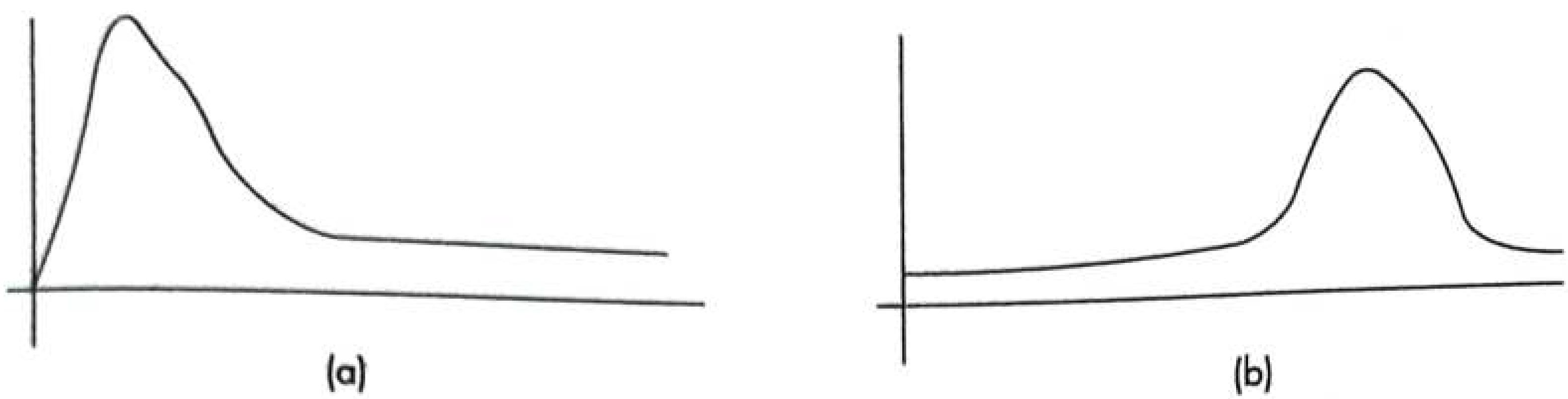

Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

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Shape

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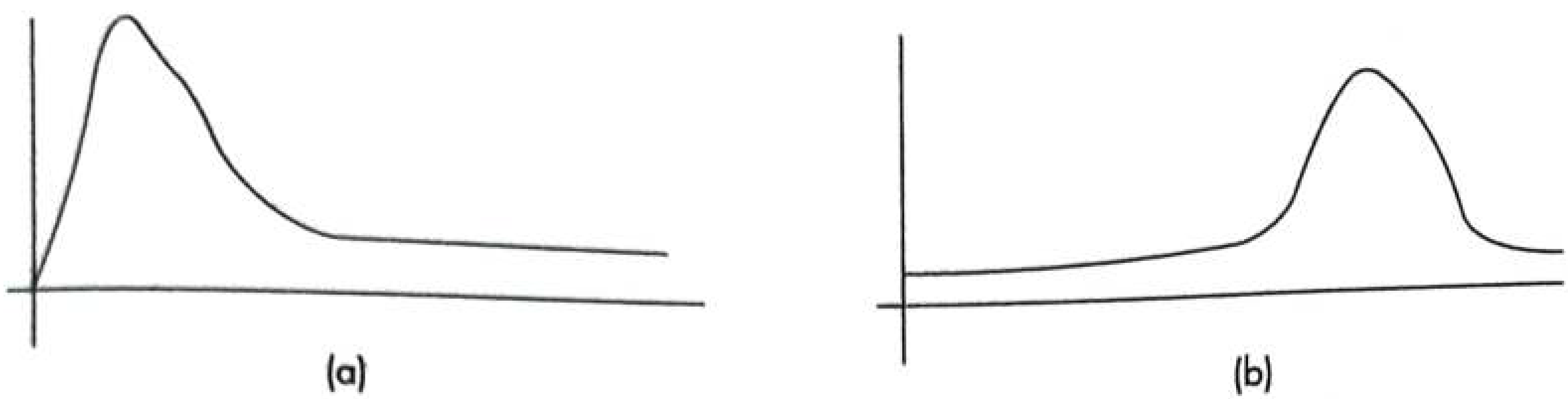

Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

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Shape

- The dataset may also either have very high values or extremely low values.
- If the dataset has far higher values, then it is said to be **skewed to the right.**
- On the other hand, if the dataset has far more low values then it is said to be **skewed towards left.**
- If the tail is longer on the left-hand side and hump on the right-hand side, it is called positive skew. Otherwise, it is called negative skew.

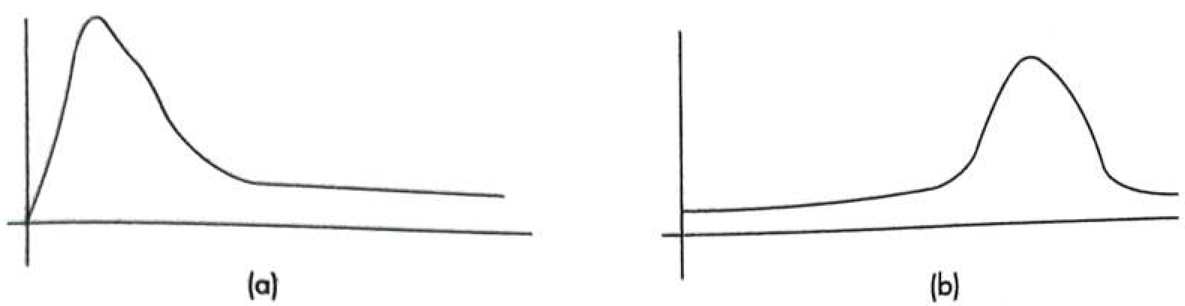

Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

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Shape

- if the data is skewed, then there is a greater chance of outliers in the dataset.
- This affects the mean and median. Hence, this may affect the performance of the data mining algorithm.
- A perfect symmetry means the skewness is zero.
- In the case of skew, the median is greater than the mean.
- In positive skew, the mean is greater than the median.
- for negatively skewed distribution, the median is more than the mean.
- The relationship between skew and the relative size of the mean and median can be summarized by a convenient numerical skew index known as Pearson 2 skewness coefficient

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Shape

- The relationship between skew and the relative size of the mean and median can be summarized by a convenient numerical skew index known **as Pearson 2 skewness coefficient**

$$\frac{3 \times (\mu - median)}{\sigma} \tag{2.12}$$

Also, the following measure is more commonly used to measure skewness. Let $X_1, X_2, \cdots, X_N$ be a set of 'N' values or observations then the skewness can be given as:

$$\frac{1}{N} \times \sum_{i=1}^{N} \frac{(x_i - \mu)^3}{\sigma^3} \tag{2.13}$$

Here, μ is the population mean and σ is the population standard deviation of the univariate data. Sometimes, for bias correction instead of N, $N - 1$ is used.

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Kurtosis

- **Kurtosis also indicates the peaks of data.**
- If the data is high peak, then it indicates higher kurtosis and vice versa.
- Kurtosis is the measure of whether the **data is heavy tailed or light tailed relative to normal distribution.**
- It can be observed that **normal distribution has bell-shaped curve with no long tails.**
- Low kurtosis tends to have light tails.
- The implication is that there is no outlier data.

$$\frac{\sum_{i=1}^{N}(x_i - \bar{x})^4/N}{\sigma^4} \tag{2.14}$$

It can be observed that $N - 1$ is used instead of N in the numerator of Eq. (2.14) for bias correction. Here, $\bar{x}$ and σ are the mean and standard deviation of the univariate data, respectively.

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Mean Absolute Deviation (MAD)

- MAD is another dispersion measure and is robust to outliers.
- Normally, the **outlier point is detected by computing the deviation from median and by dividing it by MAD.**
- Here, the **absolute deviation between the data and mean is taken.**
- Thus, the **absolute deviation** is given as:

$$|x - \mu|$$

The sum of the absolute deviations is given as $\Sigma |x - \mu|$

Therefore, the mean absolute deviation is given as: $\dfrac{\Sigma|x - \mu|}{N}$

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Coefficient of Variation (CV)

- Coefficient of variation is used to **compare datasets with different units.**
- CV is the **ratio of standard deviation and mean**, and **%CV is the percentage of coefficient of variations.**

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Special Univariate Plots

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- The ideal way **to check the shape of the dataset** is a **stem and leaf plot.**
- A stem and leaf plot are a display that help us to know the shape and distribution of the data.
- In this method, **each value is split into a 'stem' and a 'leaf.**
- The **last digit is usually the leaf and digits to the left of the leaf mostly form the stem.**

Special Univariate Plots

The stem and leaf plot for the English subject marks, say, {45, 60, 60, 80, 85} is given in Figure 2.9.

Stem	Leaf
4	5
5	
6	0 0
7	
8	0 5

Figure 2.9: Stem and Leaf Plot for English Marks

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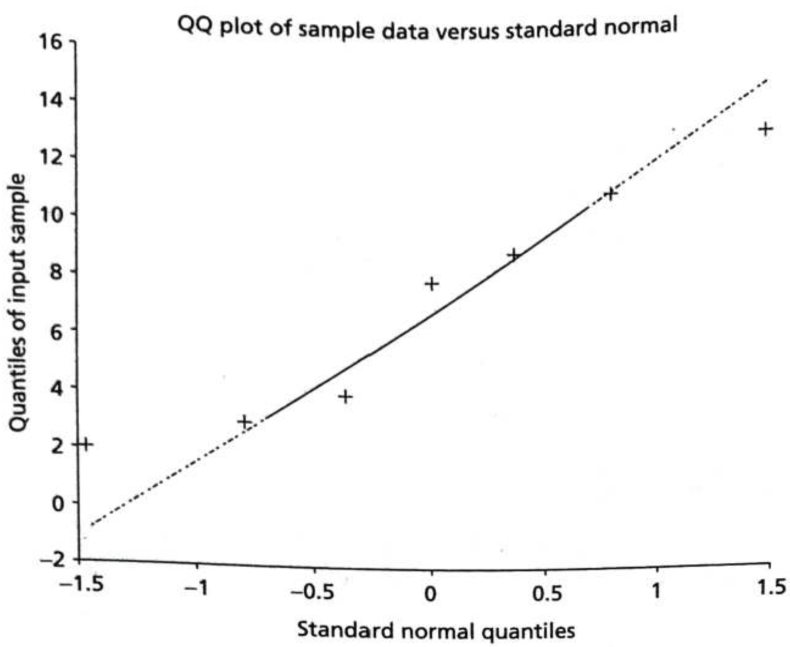
A *Q-Q* plot

- A *Q-Q* plot can be used to assess the shape of the dataset.
- The Q-Q plot is **a 2D scatter plot of an univariate data against theoretical normal distribution data or of two datasets** — the quartiles of the first and second datasets.



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